MILITARY SPECIFICATION

DISCRIMINATOR, ANTI-TANK MINE, M2 AND SPOOL ASSEMBLY

1. SCOPE

1.1 This specification covers the assembly of one type of discriminator, designated as Discriminator, Anti-Tank, Mine M2 and Spool Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

FEDERAL

FED-STD-406 - Plastics: Methods of Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1345

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9207428 - Connector, 2 Contact Female
9221336 - Discriminator, Anti-Tank, Mine: M2 and Spool Assembly
9231950 - Steel, Spring, Copper Coated
9231962 - Discriminator, AT, Mine: M2

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 9221336 - Equipment Tabulation Number

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials and parts.- Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly.- The assembly shall comply with all requirements specified on Drawing (dwg.) 9221336 and with all requirements specified in applicable specifications.

3.3 Continuity and resistance of assembly (dwg. 9231962).- The resistance of the discriminator assembly shall be five ohms, maximum, when tested as specified in 4.4.1.

3.4 Discriminator.- The discriminator shall comply with the following requirements throughout the temperature range of minus 25 degrees Fahrenheit (F.) to plus 120 degrees F.

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3.4.1 Functioning of discriminator.- The discriminator shall function when a force of 8 pounds, maximum, is applied to each of any two adjacent discriminator segments. The test shall be performed as specified in 4.4.2.

3.4.2 Non-functioning of discriminator.- The discriminator shall not function when a force of one pound, minimum, is applied to any two adjacent discriminator segments. The test shall be performed as specified in 4.4.3.

3.4.2.1 Individual segments.- The discriminator shall not function when a force of eight pounds, minimum, is applied to any active discriminator segment when tested as specified in 4.4.4.

3.4.3 Voltage breakdown.- There shall be no evidence of breakdown of insulation or closed circuits between conductors of the discriminator when tested as specified in 4.4.5.

3.5 Air leakage.- The discriminator shall show no evidence of air leakage when tested as specified in 4.4.6.

3.6 Workmanship.- All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, chips, scale, dirt, grease, rust, and other foreign material. The cleaning agent shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. There shall be no evidence of bare wires or of torn, frayed, or broken insulation.

3.7 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may

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utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Statement of findings on fungus resistance of fully compounded insulation material prior to extruding and molding and certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.- Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the Contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government

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inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 15 assemblies and 15 complete sets of parts (i.e. 15 each of every component and every sub-assembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials, (see 4.1.1e). A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.- Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

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4.3 Inspection provisions

4.3.1 Lot formation.- The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain:

a. Parts of one lot interfix number from one manufacturer.

b. Compounded insulation material from one lot from the same manufacturer.

Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examination.- Sampling plans and procedures for the following classification of defects shall be in accordance with MIL-STD-105 except that inspection for critical defects (and majors, when so specified) shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 for defects other than critical may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40

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percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Contact (see dwg. 9204869 covering a detail of dwg. 9221336).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Outside diameter	Gage	01001
102.	Knurl missing	Visual	01002
Minor:	AQL 0.65 percent		
101.	Evidence of poor workmanship (see 3.6)	Visual	01003

4.3.2.2 Cable (see dwg. 9207416 covering a detail of dwg. 9221336).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Contact cavity fails to accept gage simulating mating part (dwg. 9207428), functional	Gage	02001
102.	Length of cable, minimum (min.)	Gage	02002
103.	Wire not twisted	Visual	02003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	02004

4.3.2.3 Contact, upper (see dwg. 9231951 covering a detail of dwg. 9221336).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:	AQL 0.40 percent		
101.	Height of crown.....	Gage	03001
102.	Contact does not return to original height.....	Gage	03002
Minor:	AQL 1.50 percent		
201.	Width of contact.....	Gage	03003
202.	Thickness of contact.....	Gage	03004
203.	Evidence of poor workmanship (see 3.6).....	Visual	03005

4.3.2.4 Contact, lower (see dwg. 9231952 covering a detail of dwg. 9221336).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Height of crown.....	Gage	04001
102.	Profile of forked end.....	Gage	04002
103.	Contact does not return to original height.....	Gage	04003
Minor:	AQL 1.50 percent		
201.	Width of contact.....	Gage	04004
202.	Thickness of contact.....	Gage	04005
203.	Evidence of poor workmanship (see 3.6).....	Visual	04006

4.3.2.5 Insulation, mylar (see dwg. 9231953 covering a detail of dwg. 9221336).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		

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Minor:	AQL 1.00 percent		
201.	Improperly folded.....	Visual	05001
202.	Evidence of poor workmanship (see 3.6).....	Visual	05002

4.3.2.6 Cable, corrugated (see dwg. 9231954 covering a detail of dwg. 9221336).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Width, maximum (max.).....	Gage	06001
202.	Height of cable.....	Gage	06002
203.	Pitch of corrugations.....	Gage	06003
204.	Evidence of poor workmanship (see 3.6).....	Visual	06004

4.3.2.7 Spool assembly (see dwg. 9221337 covering a detail of dwg. 9221336).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Inside length of spool.....	Gage	07001
202.	Outside length of spool.....	Gage	07002
203.	Evidence of poor workmanship (see 3.6).....	Visual	07003

4.3.2.8 Discriminator section, exploded view, prior to assembly into outer jacket (see dwg. 9231962 covering a detail of dwg. 9221336).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:		AQL 0.65 percent	
101.	Upper contacts taped improperly.	Visual	08001
102.	Lower contacts not overlapped...	Visual	08002

Minor:		AQL 1.00 percent	
201.	Thread missing, improper or inadequate.....	Visual	08003
202.	Evidence of poor workmanship (see 3.6).....	Visual	08004

4.3.2.9 Discriminator, AT, Mine: M2 (see dwg. 9231962 covering a detail of dwg. 9221336).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Location of marked areas.....	Gage	09001
Minor:	AQL 1.00 percent		
201.	Length, max.....	Gage	09002
202.	Evidence of poor workmanship (see 3.6).....	Visual	09003

4.3.2.10 Discriminator, AT, Mine: M2 and Spool assembly (see dwg. 9221336).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Discriminator improperly coiled.	Visual	10001
202.	Pressure-sensitive tape insecure or improperly applied.....	Visual	10002
203.	Evidence of poor workmanship (see 3.6).....	Visual	10003

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4.3.3 Testing

4.3.3.1 Continuity and resistance of discriminator (see 3.3) - Major defect.- Code No. 12001.- This test shall be performed one hundred percent. Any discriminator which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2 Functioning of discriminator (see 3.4) - Major defect - Code No. 13001.- This test shall be performed one hundred percent at ambient room temperature. Any discriminator which fails to meet the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

4.3.3.2.1 Temperature conditioning at 120 degrees F. - Code No. 13002.- The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.2.1 using equipment in accordance with 4.3.4. Samples complying with the requirement shall be returned to the lot.

4.3.3.2.2 Temperature conditioning at minus 25 degrees F. - Code No. 13003.- The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.2.2 using equipment in accordance with 4.3.4. Samples complying with the requirement shall be returned to the lot.

4.3.3.3 Non-functioning of discriminator (see 3.4) - Major defect - Code No. 14001.- This test shall be performed one hundred percent at ambient room temperature. Any discriminator which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.3.

4.3.3.3.1 Temperature conditioning at 120 degrees F. - Code No. 14002.- The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.3.1 using equipment in accordance with 4.3.4. The sample items used for the test of 4.3.3.2.1 may also be used for this test. Samples complying with the requirement shall be returned to the lot.

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4.3.3.3.2 Temperature conditioning at minus 25 degrees F. - Code No. 14003.- The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.3.2 using equipment in accordance with 4.3.4. The sample items used for the test of 4.3.3.2.2 may also be used for this test. Samples complying with the requirement shall be returned to the lot.

4.3.3.4 Individual segments (see 3.4) - Major defect - Code No. 15001.- This test shall be performed one hundred percent at ambient room temperature. If any discriminator fails to comply with the applicable requirement, it shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.4.1 Temperature conditioning at 120 degrees F. - Code No. 15002.- The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.4.1 using equipment in accordance with 4.3.4. The samples used for the test of 4.3.3.2.1 may also be used for this test. Samples complying with the requirement shall be returned to the lot.

4.3.3.4.2 Temperature conditioning at minus 25 degrees F. - Code No. 15003.- The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Code Letter G and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.4.2 using equipment in accordance with 4.3.4. The sample items used for the test of 4.3.3.2.2 may also be used for this test. Samples complying with the requirement shall be returned to the lot.

4.3.3.5 Voltage breakdown (see 3.4.3) - Major defect - Code No. 16001.- This test shall be performed one hundred percent. Any discriminator which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4.

4.3.3.6 Air leakage (see 3.5) - Major defect - Code No. 17001.- The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Code Letter H and an AQL of 1.00 percent. Any discriminator which fails to comply with the applicable requirement

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shall be classed defective. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4. Samples complying with the requirement shall be returned to the lot.

4.3.3.7 Heat treatment of discriminator contact material (see dwg. 9231950) - Major defect - Code No. 18001.- A test specimen six (6) inches long shall be taken from each continuous spool of material used in the fabrication of upper and lower contacts, and tested in accordance with the heat treatment requirement as specified on dwg. 9231950. Failure of the specimen to comply with the drawing requirement shall be cause for rejection of the spool from which the specimen was taken.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 9221336 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment.- The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

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4.4 Test methods and procedures

4.4.1 Continuity and resistance of cable assembly.- The resistance of the cable assembly shall be determined while adjacent discriminator segments furthermost from the connecting wires are closed by applying a vertical force of eight pounds, minimum, through a one-half inch diameter rigid disc to each segment. The forces shall be located so that they are between eight and ten feet apart. The discriminator shall be placed on a flat horizontal surface for the performance of this test. Observation shall be made to determine compliance with the applicable requirement. Test equipment shall be in accordance with 4.3.4.

4.4.2 Functioning of discriminator.- The discriminator shall be tested by applying the specified force to each active segment of the discriminator while the contacts of the adjacent segment are closed by application of the specified force. The force shall be applied vertically through a one-half inch diameter rigid disc. The discriminator shall be placed on a flat horizontal surface for the performance of this test. The test shall be performed for each combination of adjacent segments and shall be performed on the areas of the discriminator which are marked with brown paint. Observation shall be made to determine compliance with the applicable requirement. Test equipment shall be in accordance with 4.3.4.

4.4.2.1 Temperature conditioning at 120 degrees F.- This test shall be performed as specified in 4.4.2 except that the discriminators shall be temperature conditioned on spools at 120 degrees F. plus or minus 5 degrees F. for a minimum of four hours prior to testing. The test shall be performed within four minutes after removal from the test chamber.

4.4.2.2 Temperature conditioning at minus 25 degrees F.- This test shall be performed as specified in 4.4.2.1 except that the conditioning temperature shall be minus 25 plus or minus 5 degrees.

4.4.3 Non-functioning of discriminator.- The discriminator shall be tested by applying the specified force to each active segment of the discriminator while the contacts of the adjacent segment are closed by application of the specified force. The force shall be applied vertically through a one-half inch diameter rigid disc. The discriminator shall be placed on a flat horizontal

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surface for the performance of this test. The test shall be performed for each combination of adjacent discriminator segments. Observation shall be made to determine compliance with the applicable requirement. Test equipment shall be in accordance with 4.3.4.

4.4.3.1 Temperature conditioning at 120 degrees F.- This test shall be performed as specified in 4.4.3 except that the discriminators shall be temperature conditioned on spools at 120 degrees F. plus or minus 5 degrees F. for a minimum of four hours prior to testing. The test shall be performed within four minutes after removal from the test chamber.

4.4.3.2 Temperature conditioning at minus 25 degrees F.- This test shall be performed as specified in 4.4.3.1 except that the conditioning temperature shall be minus 25 plus or minus 5 degrees.

4.4.4 Individual segments.- The discriminator shall be tested at ambient room temperature by applying the specified force to each active segment of the discriminator one at a time. The force shall be applied vertically through a one-half inch diameter rigid disc. The discriminator shall be placed on a flat horizontal surface for the performance of this test. Observation shall be made to determine compliance with the applicable requirements. Test equipment shall be in accordance with 4.3.4.

4.4.4.1 Temperature conditioning at 120 degrees F.- This test shall be performed as specified in 4.4.4 except that the discriminators shall be temperature conditioned on spools at 120 degrees F. plus or minus 5 degrees F. for a minimum of four hours prior to testing. The test shall be performed within four minutes after removal from the test chamber.

4.4.4.2 Temperature conditioning at minus 25 degrees F.- This test shall be performed as specified in 4.4.4.1 except that the conditioning temperature shall be minus 25 degrees plus or minus 5 degrees F.

4.4.5 Voltage breakdown.- A minimum electrical potential of 300 volts, direct current, shall be applied across the conductors of the discriminator for five seconds, min. The conductors shall be placed on a flat horizontal surface for the performance of this

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test. The contacts of segment no. 1 shall be closed by applying the specified force for a period of two seconds, min. while observing for compliance with the applicable requirement. The force on segment no. 1 shall be removed, then the contacts of segment no. 2 shall be closed by applying the specified force for a period of two seconds, min. again observing for compliance with the applicable requirement. Test equipment shall be in accordance with 4.3.4.

4.4.6 Air leakage.- The discriminator shall be subjected to an internal air pressure of 2 pounds per square inch (p.s.i.) plus 0.5 p.s.i., minus zero p.s.i. for a minimum of one minute while submerged under water. Observation shall be made for a steady stream of bubbles to determine evidence of leakage. Test equipment shall be in accordance with 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 The discriminator and spool assembly shall be packed and marked in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Submission of first article quantity.-Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out, may be obtained from the contracting officer.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

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6.3.1 Equipment Lists.-Equipment part or identifying lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 through 6.3.6. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.5 and 6.3.6 will be designated in the EL by the following codes. Other inspection equipment listed on the EL and not coded are defined in 6.3.4.

MU - Army design, mandatory for use.
CD - Contractor design and/or commercial equipment on contractor format. (Requires submission for Government approval as specified in 6.3.7).

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.4 Envelope drawing.-Envelope drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of envelope drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of envelope drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility

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shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data and ammunition data cards.- In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-ND5
SMUPA-DC

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1345-A-235

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FOLD

DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 114-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT. 61 1426

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